6. Acid rain is formed by sulfur dioxide gas from industrial processes escaping into the atmosphere and reacting with water in clouds.

In recent years, scientists have developed sulfur scrubbers to stop sulfur dioxide gas escaping into the atmosphere from coal-fired power plants. The scrubbers are placed in the chimneys and trap the sulfur dioxide.

There are two types of scrubbers – wet scrubbers and dry scrubbers.

Wet scrubbers

Water is sprayed down the chimneys onto beds of crushed limestone. Sulfur dioxide is absorbed by the water forming an acidic solution which is neutralised by the limestone.

Wet scrubbing can be used in small and large power plants. During wet scrubbing 4% of sulfur dioxide escapes.

Dry scrubbers

A mixture of dry alkaline chemicals is sprayed into the chimneys. Some of the dry chemicals neutralise the sulfur dioxide.

Dry scrubbing is limited to small or medium sized power plants. No water is used so costs are lower. During dry scrubbing 10% of sulfur dioxide escapes.

Examiner only

(a) Tick (✓) the physical change happening to the sulfur dioxide in a wet scrubber. [1]

it freezes ☐

it dissolves ☐

it condenses ☐

it melts ☐

(b) Tick (✓) the pH change that happens as a solution of sulfur dioxide is neutralised in a wet scrubber. [1]

pH 11 to pH 7 ☐

pH 4 to pH 7 ☐

pH 7 to pH 11 ☐

pH 7 to pH 4 ☐

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(c) The table shows some statements about wet and dry scrubbing.

Complete the table using a tick (✓) to show whether each statement applies to wet scrubbing only, to dry scrubbing only or to both wet and dry scrubbing. [3]

Statement	Wet scrubbing only	Dry scrubbing only	Both wet and dry scrubbing
Can be used in large power plants			
At least 90% efficient			
Neutralises sulfur dioxide			

Examiner only

(d) The table shows the mass of sulfur dioxide released into the atmosphere per year in the UK every five years between 1990 and 2015.

Year	Mass of sulfur dioxide released (millions of tonnes)
1990	3.50
1995	0.60
2000	0.40
2005	0.35
2010	0.30
2015	0.20

Describe the trend in the mass of sulfur dioxide released into the atmosphere between 1990 and 2015. [2]

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